Undergradute Project - 2 Presentation

A Comparative Study of First Order and WENO Schemes for solving Hughes' Crowd Dynamics Model

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Crowd Dynamics

Fluid Dynamics	Crowd Dynamics
Fluid Dynamics studies about the flow of fluid. Phenomenons like how would the fluid behaves or interact in the domain.	Crowd Dynamics studies about the flow of crowd.
Fluid mechanics has a macroscopic model, like the navier strokes equation and also microscopic models.	Crowd Dynamics also has many model, soem being macroscopic and some being microscopic.

This project uses a Macroscopic model known as Hughes' Model .

Hughes’ Model

- Hughes introduces the concept of potential. Potential in a problem is a scalar field which is the tiem taken by pedestriana t any point ont he domain to reach the exit.
- So the model now has two unknowns and they are Density and Potential, that are required to eb solved at each time step.

These are the assumptions made by Hughes’ to define the Model.

Assumption 1

Speed of the pedestrian is determined by the local density around them.

The relation chosen for this project is **Green Sheild’s Relation**

$\rho(x,y,t)$: DENSITY

$U(\rho)$: SPEED

$$U(\rho) = U_{Max}\left[1 - \frac{\rho}{\rho_{Max}}\right]$$

Assumption 2

A potential Field exist such that pedestrian move right angle to the line of constant potential to minimize it.

$g(\rho)$: DISCOMFORT

$U(\rho)$: SPEED

ϕ : POTENTIAL

$$\frac{1}{\|\nabla\phi\|} = U(\rho) \cdot g(\rho)$$

Assumption 3

The rate of increase in travel time with distance is inversely proportional to walking speed and discomfort function.

U : SPEED

ϕ : POTENTIAL

u : X-COMPONENT

v : Y-COMPONENT

$$u = \frac{-\frac{\partial\phi}{\partial x}}{\|\nabla\phi\|} * U(\rho)$$

$$v = \frac{-\frac{\partial\phi}{\partial y}}{\|\nabla\phi\|} * U(\rho)$$

Assumption 4

The number of pedestrians are conserved.

u : X COMPONENT VELOCITY

v : Y COMPONENT VELOCITY

ρ : DENSITY

$$\frac{\partial\rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0$$

Algorithm Flow

Initialize the Density (Initial Condition)
Density at each point has to be specified at t=0

Now the density is known at each point at this time step

Speeds at every point is found
And this is done using Greenshields's method

Potential at every point is obtained
Using Speed & Discomfort at each point along Fast Sweep, the potential at each point is obtained

Velocity vector at every point is obtained
Using Speed at each point (Magnitude), and Potential Field (direction), velocity vector field is obtained

Density at every point at next time step is obtained
Using the velocity at each time step, and density at the current time step the value of density at next time step is evaluated, using Lax Wendroff Scheme

I have used Finite Differnce Method to solve the Governing equation and there are two schemes that have been tested to solve the Eikonal and Advection Equation.

First Order Scheme

I had used the First Order Fast Sweep Method to solve the Eikonal Equation and the stencil for the same is given below.

Given equation: $|\nabla\phi(x)| = f(x)$ on Ω

$\phi(x) = 0$ on Γ_{exit}

$\frac{\partial\phi}{\partial n} = 0$ on Γ_{wall}

$$\phi^{x}_{Min} = Min(\phi^{(n)}_{i-1,j}, \phi^{(n)}_{i+1,j})$$

$$\phi^{y}_{Min} = Min(\phi^{(n)}_{i,j-1}, \phi^{(n)}_{i,j+1})$$

$$\phi^{(n+1)}_{ij} = \begin{cases} Min(\phi^{x}_{Min}, \phi^{y}_{Min}) + f_{ij}h \; ; & |\phi^{x}_{Min} - \phi^{y}_{Min}| \geq f_{ij}h \\ \frac{\phi^{x}_{Min} + \phi^{y}_{Min} + (2f_{ij}^{2}h^{2} - (\phi^{x}_{Min} - \phi^{y}_{Min})^{2})^{1/2}}{2} \; ; & |\phi^{x}_{Min} - \phi^{y}_{Min}| \leq f_{ij}h \end{cases}$$

I had used the Lax Wendroff Scheme to solve the Advection Equation and the stencil for the same is given below.

Lax Wendroff scheme:

$$\rho^{n+1/2}_{j+1/2} = 1/2(\rho^{n}_{j+1} + \rho^{n}_{j}) - \frac{\Delta t}{2\Delta x}(F^{n}_{j+1} - F^{n}_{j})$$

$$\rho^{n+1/2}_{j-1/2} = 1/2(\rho^{n}_{j} + \rho^{n}_{j-1}) - \frac{\Delta t}{2\Delta x}(F^{n}_{j} - F^{n}_{j-1})$$

$$\rho^{n+1}_{j} = \rho^{n}_{j} - \left(\frac{\Delta t}{\Delta x}\right)(F^{n+1/2}_{j+1/2} - F^{n+1/2}_{j-1/2})$$

~~And~~ where, $F^{n}_{j} = f(\rho^{n}_{j}) \cdot \rho^{n}_{j} = (1 - \rho^{n}_{j}) \cdot \rho^{n}_{j}$

- We initialize φ with infinity, then in each sweep, every point looks at its neighbors and updates its value to the minimum neighbor value plus the tiem taken to move at that point.

Boundary Condition is that the value of the density at the walls is specifiied and the value of potential is equal to zero at the exit, and the normal component of potiential is equal to zero at the walls.

Weighted Essentially Non-Oscillatory Scheme

I had used the third order WENO Fast Sweep to solve the Eikonal Equation and the stencil for the same is given below.

$$\phi_{i,j}^{\text{new}} = \begin{cases} \min(\phi_{i,j}^{x\min}, \phi_{i,j}^{y\min}) + c_{i,j}h, & \text{if } |\phi_{i,j}^{x\min} - \phi_{i,j}^{y\min}| \leqslant c_{i,j}h, \\ \dfrac{\phi_{i,j}^{x\min} + \phi_{i,j}^{y\min} + (2c_{i,j}^2h^2 - (\phi_{i,j}^{x\min} - \phi_{i,j}^{y\min})^2)^{\frac{1}{2}}}{2}, & \text{otherwise,} \end{cases}$$

where $c_{i,j} = C(x_i, y_j, t)$, and

$$\begin{cases} \phi_{i,j}^{x\min} = \min(\phi_{i,j}^{\text{old}} - h(\phi_x)_{i,j}^-, \phi_{i,j}^{\text{old}} + h(\phi_x)_{i,j}^+), \\ \phi_{i,j}^{y\min} = \min(\phi_{i,j}^{\text{old}} - h(\phi_y)_{i,j}^-, \phi_{i,j}^{\text{old}} + h(\phi_y)_{i,j}^+), \end{cases}$$

with

$$(\phi_x)_{ij}^- = (1 - w_-)\left(\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2h}\right) + w_-\left(\frac{3\phi_{ij} - 4\phi_{i-1,j} + \phi_{i-2,j}}{2h}\right),$$

$$(\phi_x)_{ij}^+ = (1 - w_+)\left(\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2h}\right) + w_+\left(\frac{-3\phi_{ij} + 4\phi_{i+1,j} - \phi_{i+2,j}}{2h}\right),$$

$$w_- = \frac{1}{1 + 2r_-^2}, \quad r_- = \frac{\varepsilon + (\phi_{ij} - 2\phi_{i-1,j} + \phi_{i-2,j})^2}{\varepsilon + (\phi_{i+1,j} - 2\phi_{ij} + \phi_{i-1,j})^2},$$

$$w_+ = \frac{1}{1 + 2r_+^2}, \quad r_+ = \frac{\varepsilon + (\phi_{ij} - 2\phi_{i+1,j} + \phi_{i+2,j})^2}{\varepsilon + (\phi_{i+1,j} - 2\phi_{ij} + \phi_{i-1,j})^2}.$$

I had used the fifth order WENO Advection Scheme to solve the Advection Equation and the stencil for the same is given below.

$$\frac{d}{dt}\rho_{ij} = -\frac{1}{\Delta x}\left((\hat{f}_1)_{i+\frac{1}{2}j} - (\hat{f}_1)_{i-\frac{1}{2}j}\right) - \frac{1}{\Delta y}\left((\hat{f}_2)_{i,j+\frac{1}{2}} - (\hat{f}_2)_{i,j-\frac{1}{2}}\right),$$

$$f_1^{\pm}(\rho) = \frac{1}{2}(f_1(\rho) \pm \alpha\rho),$$

$s_\ell = f_1(x_\ell, y_j, t)$ for $\ell = i-2, i-1, \ldots, i+2$.

$$\begin{aligned} \hat{s}_{i+\frac{1}{2}}^{(1)} &= \tfrac{1}{3}s_{i-2} - \tfrac{7}{6}s_{i-1} + \tfrac{11}{6}s_i, \\ \hat{s}_{i+\frac{1}{2}}^{(2)} &= -\tfrac{1}{6}s_{i-1} + \tfrac{5}{6}s_i + \tfrac{1}{3}s_{i+1}, \\ \hat{s}_{i+\frac{1}{2}}^{(3)} &= \tfrac{1}{3}s_i + \tfrac{5}{6}s_{i+1} - \tfrac{1}{6}s_{i+2}. \end{aligned} \tag{31}$$

We then define the three nonlinear weights ω_m for $m = 1, 2, 3$, by

$$\omega_m = \frac{\tilde{\omega}_m}{\sum_{l=1}^3 \tilde{\omega}_l}, \quad \tilde{\omega}_l = \frac{\gamma_l}{(\varepsilon + \beta_l)^2}, \tag{32}$$

where the linear weights γ_l are given by

$$\gamma_1 = \frac{1}{10}, \quad \gamma_2 = \frac{3}{5}, \quad \gamma_3 = \frac{3}{10}, \tag{33}$$

and the smoothness indicators β_l, which measure the smoothness of the approximation in the lth substencil, are given by

$$\begin{aligned} \beta_1 &= \frac{13}{12}(s_{i-2} - 2s_{i-1} + s_i)^2 + \frac{1}{4}(s_{i-2} - 4s_{i-1} + 3s_i)^2, \\ \beta_2 &= \frac{13}{12}(s_{i-1} - 2s_i + s_{i+1})^2 + \frac{1}{4}(s_{i-1} - s_{i+1})^2, \\ \beta_3 &= \frac{13}{12}(s_i - 2s_{i+1} + s_{i+2})^2 + \frac{1}{4}(3s_i - 4s_{i+1} + s_{i+2})^2. \end{aligned} \tag{34}$$

$$f_1(\rho) = f_1^+(\rho) + f_1^-(\rho)$$

$$(\hat{f}_1)_{i+\frac{1}{2}j} = \omega_1\hat{s}_{i+\frac{1}{2}}^{(1)} + \omega_2\hat{s}_{i+\frac{1}{2}}^{(2)} + \omega_3\hat{s}_{i+\frac{1}{2}}^{(3)}$$

- So WENO allows to change the order of scheme depending on which poitn in the domian are we trying to analyze.
- So regions where the solution is smooth the scheme becomes higher order (and it does that by adjusting the weights), and in regions where there are high shocks the order of stencil used is low to avoid gibbs oscillations.
- Also the WENO Advection will be used to obtain the semi descritized form of the equation , however the time is descrtized using the third order TVD Runga Kutta Method.

Boundary Condition is that at walls the values of potentials are extrapolated, and the values of potential at the exit is equal to zero. And values of density

Problem Description

A diagram of problem is given below

Inflow Condition :

$$f_1(0,y,t) = \begin{cases} t/12 & 0 \leqslant t \leqslant 60, \\ 10 - t/12 & 60 \leqslant t \leqslant 120, \\ 0 & 120 \leqslant t \leqslant 0, \end{cases}$$

$$f_2(0,y,t) = 0 \quad \forall y \in (0,50), \quad t \in (0,300),$$

Velocity Relation :

$$U(\rho) := 2(1 - \rho/10),$$

Discomfort Relation :

$$G(\rho) := 0.002\rho^2,$$

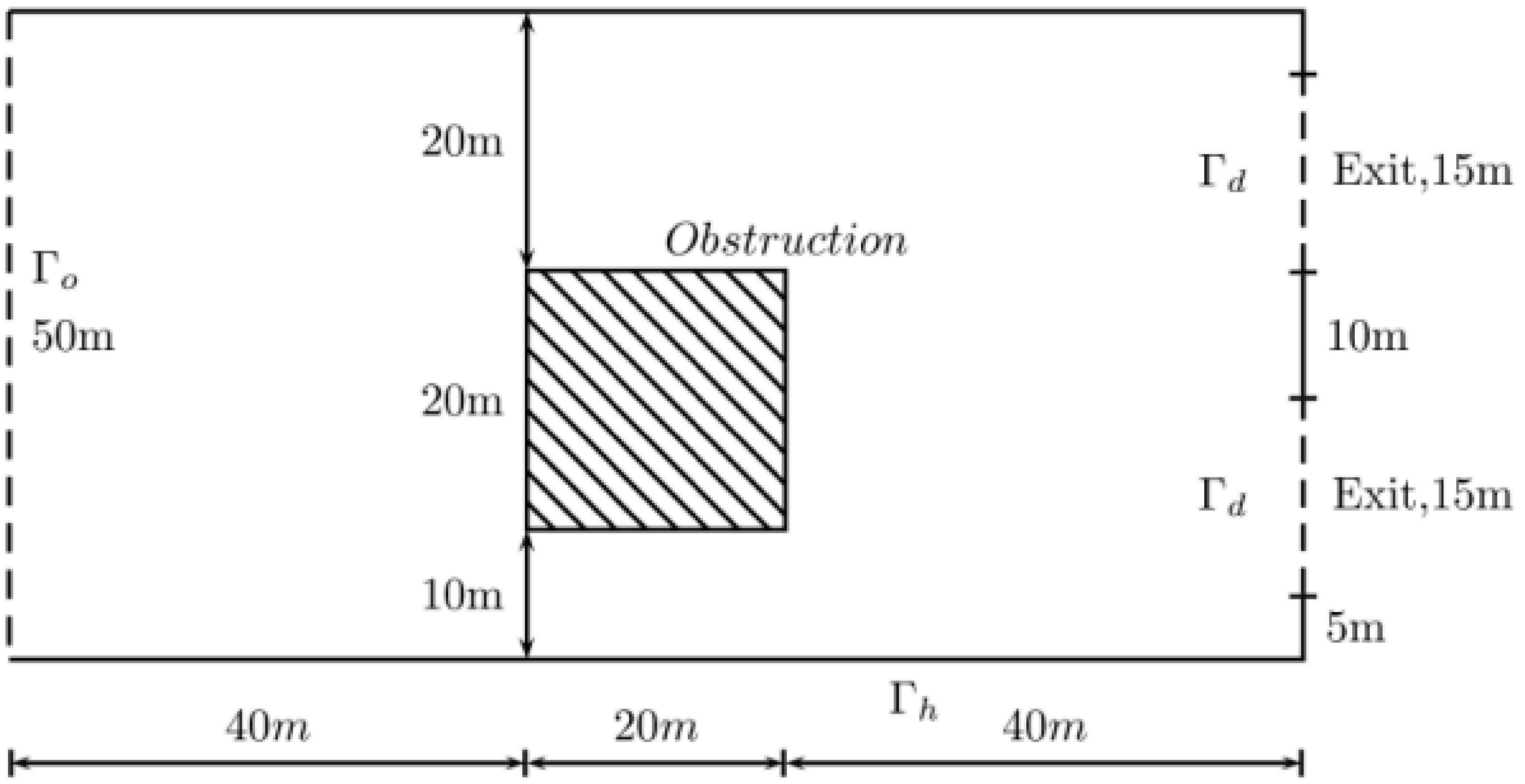
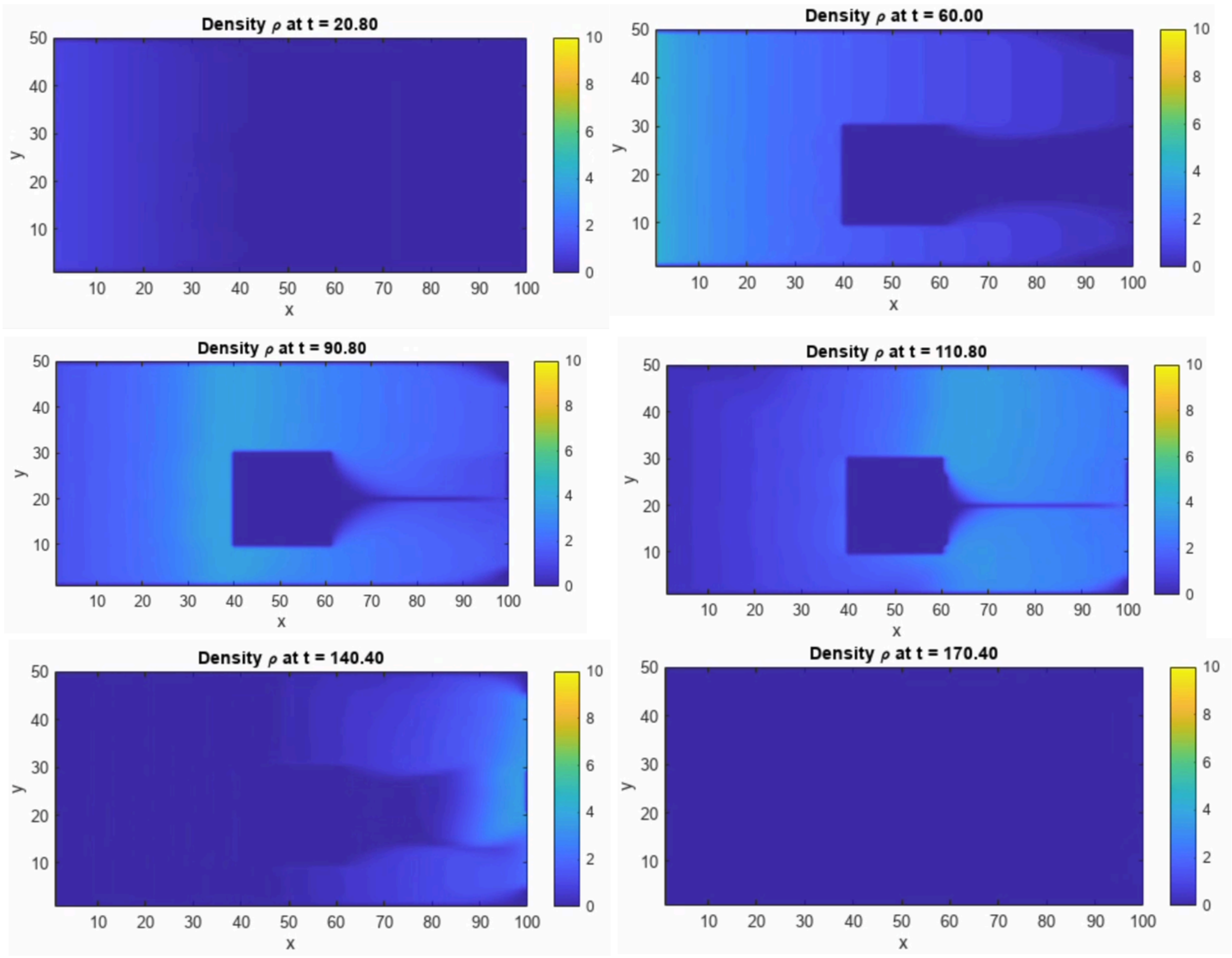


Fig. 1. The geometry of the railway platform.

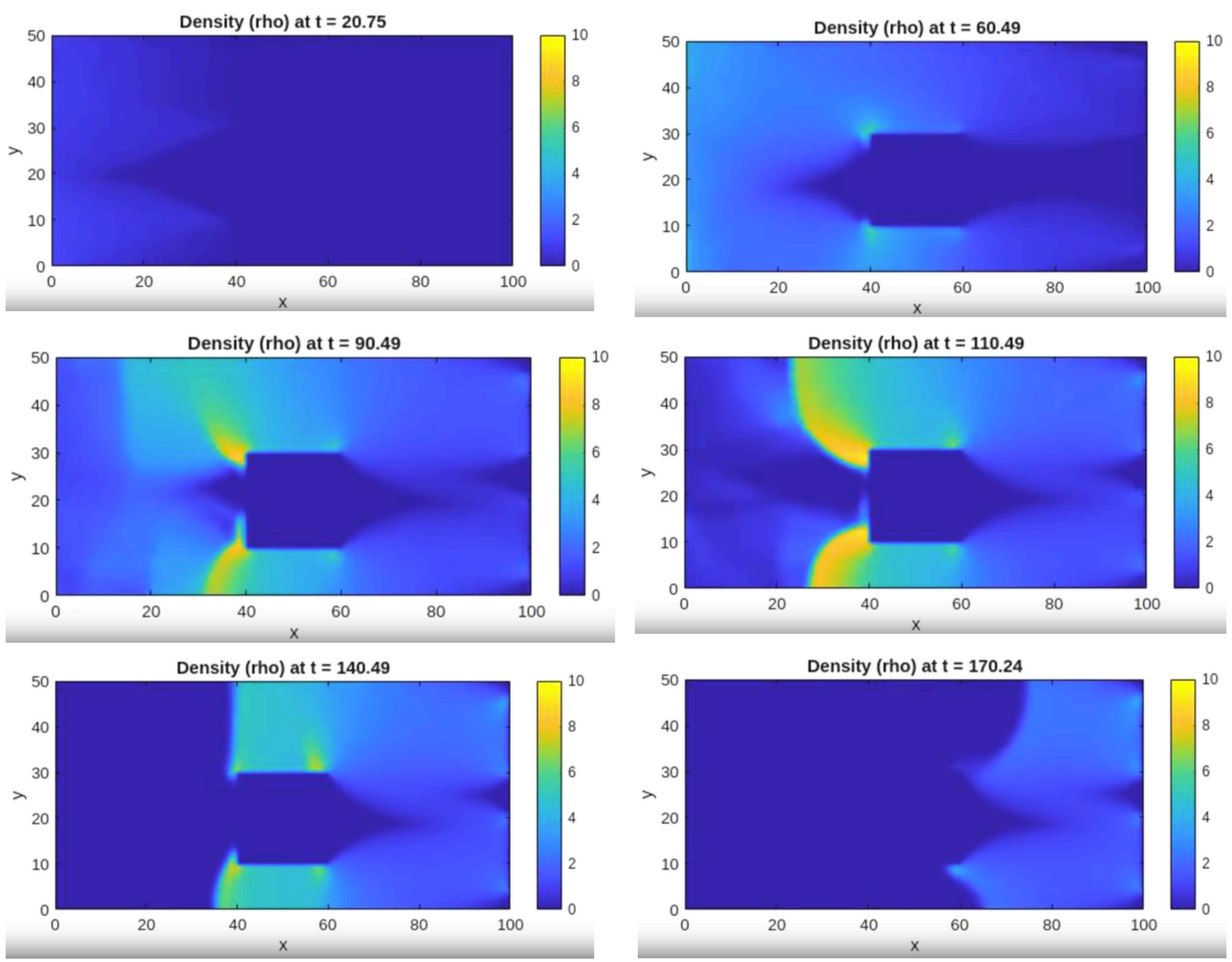
The values of density inside the obstacle is kept to be equal to zero. And similarly, the value of potential is kept equal to infinity at points inside the obstacle. Also no computation happens inside the obstacle. The condiiton at wall is discussed in the previous slides.

Results

First Order Scheme :



WENO Scheme :



It is clear from above that the WENO scheme is able to capture the Shocks much better than the First order scheme. And apart from that there are soem observation :

- Shocks are observed due to the greensheild relation which says velocity of crowd will be inversly proportional to density. (High density crowd ahead will move slower than low density crowd behind). However the shpae of the shock is still a topic of discussion.
- Another observation is that the crowd is much fatser in the case of WENO compared to firts order. This could be because of shock formation in WENO scheme that causes the crowd to slow down. Also this might be because of the diffrence in the boundary condition on density in both the schemes.

Reference

- Revisiting Hughes' dynamic continuum model for pedestrian flow and the development of an efficient solution algorithm (Ling Huang, S.C. Wong, Mengping Zhang, Chi-Wang Shu, William H.K. Lam)
- A continuum theory for the flow of pedestrians (Roger L. Hughes)